

SOFTWARE ENGINEERING LAB

Experiment No. 4: Project-Based Data Flow Diagram (DFD)

Use your own selected software project

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Class/Section	MCA	Date	03 September 2026

1. Aim

To prepare a Context Diagram, Level 0 DFD and one Level 1 DFD for your own software project.

2. Your Project Details

Project Title	TeamFinder
One-line purpose of the project	Helps students find compatible teammates for projects and hackathons based on skills and interests.
Main users/external systems	Guest, Student, Admin
Three to six major functions	Registration & login; Profile & skills; Find teammates; Projects & hackathons; Collaboration requests.
Earlier project document used (requirements/use case, if available)	TeamFinder SRS (Lab 1) and TeamFinder Use Case Diagram.

3. Before Drawing the DFD

S. No.	External Entity	Data sent to system	Data received from system
1	Guest	Registration details; Login details	Registration / login result
2	Student	Login details; Profile & skills; Search request; Project data; Collaboration request	Login status; Updated profile; Matched teammates; Project information; Request status
3	Admin	User / project management data; Report request	User information; Project information; Reports / analytics

4. Major Processes of Project

Process No.	Process Name	What does this process do?
1	Registration & Login	Registers users and verifies login credentials.
2	Manage Profile & Skills	Creates/updates profile and skills.
3	Find Teammates	Searches and matches teammates using skills and branch.
4	Projects & Hackathons	Manages project and hackathon information.
5	Collaboration Requests	Sends and tracks collaboration requests.

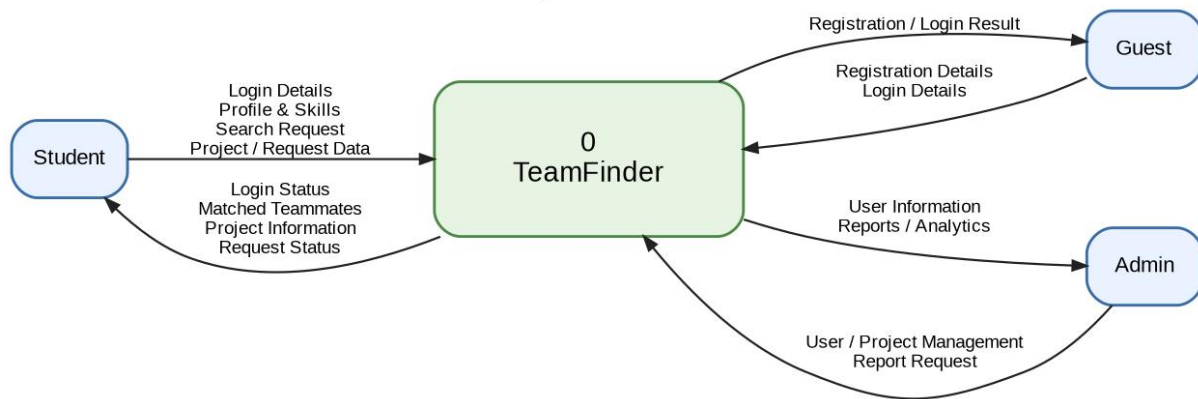
5. Data Stores

Important places where our project stores data.

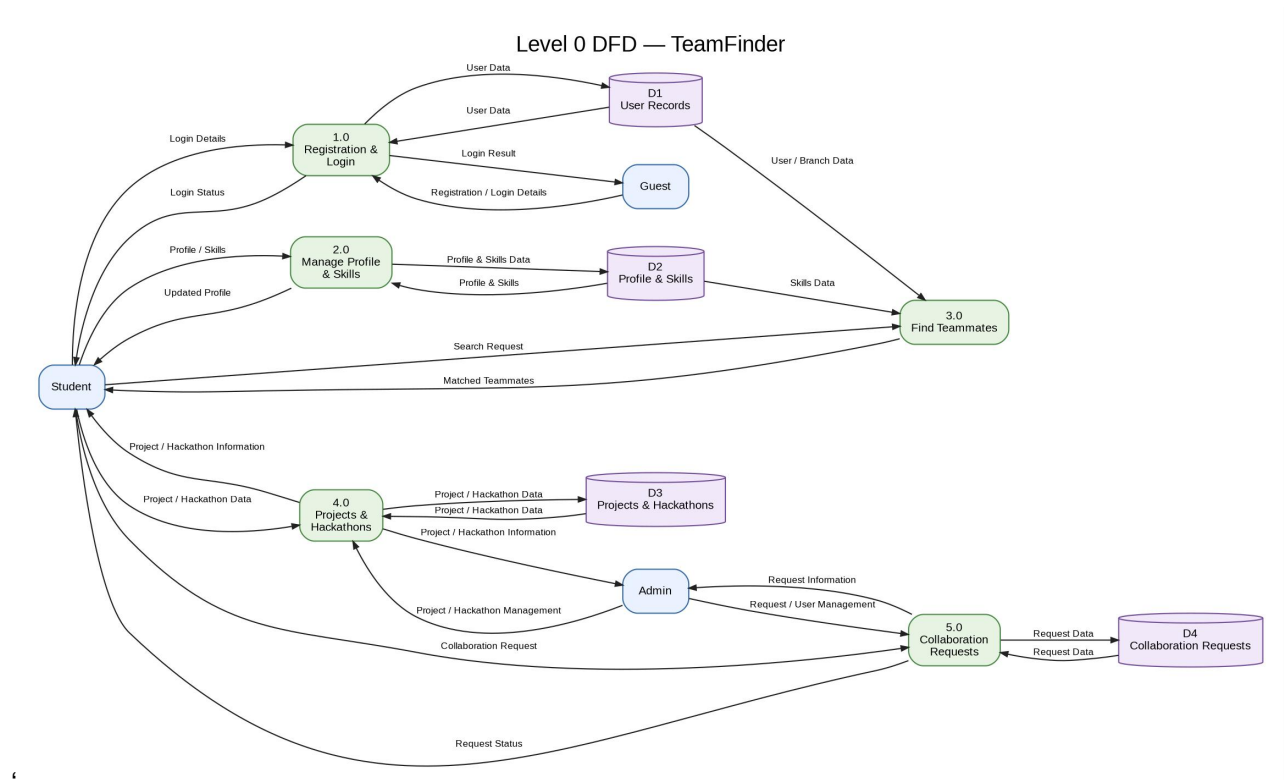
Data Store	Name	What data is stored?
D1	User Records	User account, login and branch information.
D2	Profile & Skills	Profile details, skills and interests.
D3	Projects & Hackathons	Project and hackathon information.
D4	Collaboration Requests	Request details and request status.

6. Context Diagram

Context Diagram — TeamFinder



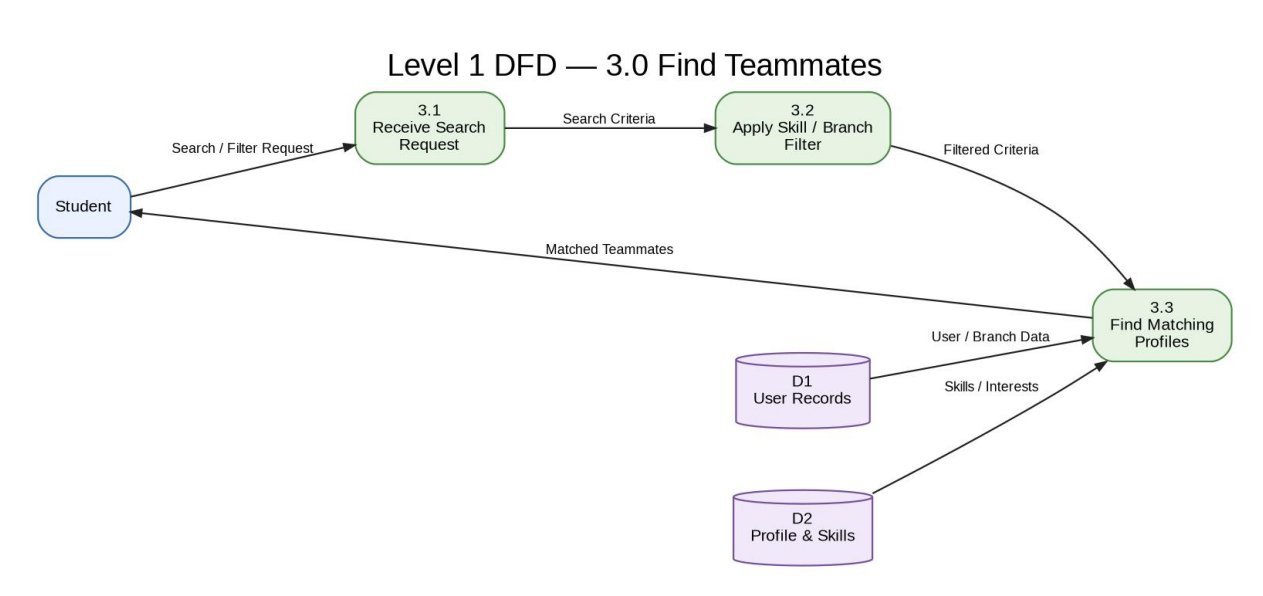
7. Level 0 DFD



8. Level 1 DFD

Choose ONE process from your Level 0 DFD and break it into smaller steps.

Selected Level 0 Process	3.0 Find Teammates
Why did you select it?	It is a main TeamFinder function and directly supports teammate search and skill-based matching.
Smaller sub-processes	3.1 Receive Search Request; 3.2 Apply Skill / Branch Filter; 3.3 Find Matching Profiles.



9. Balancing Check

Check	Yes/No	Correction made, if any
Context Diagram and Level 0 are consistent.	Yes	Main Student inputs/outputs remain consistent between Context Diagram and Level 0.
Selected Level 0 process and Level 1 are consistent.	Yes	Process 3.0 keeps Search / Filter Request input and Matched Teammates output in Level 1.
No unrelated data suddenly appears in Level 1.	Yes	No unrelated external data is introduced in Level 1.

10. Result

The DFD for TeamFinder was prepared using the project requirements and use case diagram. The Context Diagram, Level 0 and Level 1 DFD show the main data flow in the system.